

# THE OPTIMAL MONETARY POLICY RESPONSE TO TARIFFS

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# Motivation

- **How should a central bank respond to import tariffs?**
  - ▶ Tighten monetary policy to contain inflationary pressures, or...
  - ▶ Neutral monetary stance (“look-through”) and allow a one-time jump in the CPI?

## Jay Powell pushes back on calls for Federal Reserve rate cuts as soon as July

US central bank chair tells congressional committee economy remains 'solid' but tariffs could push up prices



Jay Powell has been under fire from the US president over the Federal Open Market Committee's decision to keep interest rates on hold © Mark Schiefelbein/AP



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Governor Chris Waller says US has yet to see an inflation 'shock' from Donald Trump's tariffs



Christopher Waller joined the Fed's policy-setting panel in 2020 after being nominated by Donald Trump during his first term as president © Bloomberg

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## This paper:

- ▶ Optimal monetary policy response to tariffs is **expansionary**

# Overview

- Open-economy New Keynesian model with home and imported goods
  - Macroeconomic effects depend on monetary policy

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- Tariffs can lead to an expansion or contraction in output
- $\neq$  textbook cost-push shock

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- ↓ Trade deficit, even with permanent tariff
- Monetary stimulus leads to temporary rise in output and savings

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- Exchange-rate depreciation, unlike conventional view

→ Weak dollar since April 2nd

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Exchange-rate depreciation, unlike conventional view

- Extensions: temporary/anticipated, ex/endogenous TOT, supply chains

# Literature on Tariffs in International Macro

- Classic question: Are tariffs expansionary or contractionary? Keynes vs. Mundell
- Recent studies: Auray, Devereux, Eyquem (2022,2024); Eichengreen (2019); Barattieri, Cacciatore and Ghironi (2021); Comin and Johnson (2021); Jeanne (2021); Jeanne and Son (2024); Bergin and Corsetti (2021); Erceg, Prestipino and Raffo (2023); Lloyd and Marin (2024)
- Bergin-Corsetti (2023): under cooperation, *contractionary* policy for tariff-imposing country

## OUR CONTRIBUTION:

- *Expansionary* policy is optimal. Fiscal externality  $\Rightarrow$  tariff  $\neq$  TOT shock
- Analytical results: contractionary/expansionary effects through labor supply channel
- New predictions for exchange rates and trade deficits

**Active agenda!**

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- No market power in goods or capital markets:
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  - Monetary policy does not affect terms of trade
- Monetary authority: sets monetary policy optimally, taking as given tariffs  $\{\tau_t\}$

# Households

$$\sum_{t=0}^{\infty} \beta^t \left[ U(c_t^h, c_t^f) - v(\ell_t) \right]$$

$$U(c_t^h, c_t^f) = \frac{\sigma}{\sigma-1} \left[ \omega (c_t^h)^{1-\frac{1}{\gamma}} + (1-\omega) (c_t^f)^{1-\frac{1}{\gamma}} \right]^{\frac{\gamma}{\gamma-1} \frac{\sigma-1}{\sigma}}, \quad v(\ell_t) = \omega \frac{\ell_t^{1+\psi}}{1+\psi}$$

- Budget constraint:

$$P_t^h c_t^h + P_t^f (1 + \tau_t) c_t^f + \frac{e_t b_{t+1}}{R^*} + \frac{B_{t+1}}{R_t} = e_t b_t + B_t + W_t \ell_t + T_t + D_t$$

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- Law of one price (before tariffs):  $P_t^h = e_t P_t^{h*}, \quad P_t^f = e_t P_t^{f*}$

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- Law of one price (before tariffs):  $P_t^h = e_t P_t^{h*}, \quad P_t^f = e_t P_t^{f*}$
- Terms-of-trade exogenous  $p \equiv \frac{P_t^{f*}}{P_t^{h*}} \quad \Leftarrow \text{Limit case w/ export elasticity} = \infty$

## Firms

- Production of final home good is competitive

$$Y_t = \left( \int_0^1 y_{jt}^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}}$$

- Intermediate good varieties

$$y_{jt} = \ell_{jt}$$

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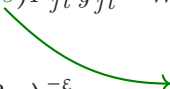
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- ▶ Monop. competitive w/ Rotemberg price adjustment costs  $\varphi$

$$\begin{aligned} \max_{\{y_{jt}, P_{jt}\}} \sum_{t=0}^{\infty} \Lambda_{t+1} & \left[ (1 + s) P_{jt} y_{jt} - W_t y_{jt} - \frac{\varphi}{2} \left( \frac{P_{jt}}{P_{j,t-1}} - 1 \right)^2 P_t^h y_t \right] \\ \text{s.t.} \quad y_{jt} &= \left( \frac{P_{jt}}{P_t^h} \right)^{-\varepsilon} y_t \end{aligned}$$


 Constant subsidy to correct markup distortion

## Firms

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### ► NK Phillips Curve

$$(1 + \pi_t) \pi_t = \frac{\varepsilon}{\varphi} \left[ \frac{W_t}{P_t^h} - 1 \right] + \beta \frac{u_h(c_{t+1}^h, c_{t+1}^f)}{u_h(c_t^h, c_t^f)} \frac{\ell_{t+1}}{\ell_t} (1 + \pi_{t+1}) \pi_{t+1}$$

where  $\pi_t \equiv P_t^h / P_{t-1}^h - 1$  is Producer Price Index (PPI) inflation

# Competitive Equilibrium

- Optimization (households and firms) + govt. budget + labor market clearing.


$$\tau_t P_t^f c_t^f = T_t + s P_t^h y_t$$



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- Balance of payments:

$$\underbrace{\left(1 - \Upsilon \frac{\varphi}{2} \pi_t^2\right) \ell_t - c_t^h}_{\text{exports}} - \underbrace{p c_t^f}_{\text{imports}} = \underbrace{\frac{b_{t+1}}{R^*} - b_t}_{\text{capital outflows}}$$

➤ Fraction of price adjustment costs that are a deadweight loss ( $1 - \Upsilon$  rebated)



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- Portfolio undetermined, assume  $B_0 = 0$      $\Leftarrow$  Abstract from valuation effects

# Efficient Allocation

$$\begin{aligned} \max_{\{b_{t+1}, c_t^f, c_t^h, \ell_t\}} \quad & \sum_{t=0}^{\infty} \beta^t [u(c_t^h, c_t^f) - v(\ell_t)], \\ \text{s.t.} \quad & c_t^h + pc_t^f + \frac{b_{t+1}}{R^*} = b_t + \ell_t. \end{aligned}$$

## EFFICIENT ALLOCATION

$$\frac{v'(\ell_t)}{u_h(c_t^h, c_t^f)} = 1$$

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 $\frac{W}{P_h} = \frac{v'(\ell_t)}{u_h(c^h, c^f)}$

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- **Tariffs:** distort  $MRS = p$  condition
- **Sticky prices:** labor wedge & inflation costs

} Two distortions

## COMPETITIVE EQUILIBRIUM $\tau = 0$

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# COMPETITIVE EQUILIBRIUM $\tau = 0$ (with $\pi_t = 0$ )

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# Employment under Look-Through Policy

**Definition:** A policy of “look-through” targets PPI inflation,  $\pi_t = 0$  for all  $t$

- Closes labor wedge and replicates flex-price allocation

→ Absent tariffs, this is optimal  $\Leftarrow$  Divine coincidence

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**Proposition.** Assume that  $\beta R^* = 1, \tau_t = \tau, b_0 = 0$ . Then, employment is given by

$$\ell_t(\tau) = \left[ \frac{\Theta_\tau + \tau}{1 + \tau} (\omega \Theta_\tau)^{\frac{\sigma - \gamma}{\gamma - 1}} \right]^{\frac{1}{1 + \sigma \psi}}, \quad \text{where } \Theta_\tau \equiv 1 + \left( \frac{1 - \omega}{\omega} \right)^\gamma [p(1 + \tau)]^{1 - \gamma} > 1$$

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and

$$c_t^h(\tau) = \frac{1 + \tau}{\Theta_\tau + \tau} \ell_t(\tau), \quad c_t^f(\tau) = \frac{\Theta_\tau - 1}{p(\Theta_\tau + \tau)} \ell_t(\tau)$$

# Are Tariffs Expansionary or Contractionary?

- Under look-through policy

$$\frac{d \log \ell(\tau)}{d\tau} = \overbrace{-\frac{(\Theta_\tau - 1)}{(1 + \sigma\psi)(1 + \tau)(\Theta_\tau + \tau)\Theta_\tau}}^{<0} [\sigma\Theta_\tau + (\sigma - \gamma)\tau]$$

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Two changes in relative prices:

- Price between home good and foreign good
- Price between labor and foreign good



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- For small  $\tau$ , increase in tariffs are always contractionary—even in the absence of TOT or exchange rate movements (cf. Mundell, 1961)

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- ▶ For small  $\tau$ , increase in tariffs are always contractionary—even in the absence of TOT or exchange rate movements (cf. Mundell, 1961)
- ▶ For large  $\tau$ : always contractionary if goods are Hicksian complements ( $\sigma > \gamma$ )
  - But may be expansionary if goods are Hicksian substitutes ( $\sigma < \gamma$ )

# Ramsey Optimal Monetary Policy

$$\max_{\pi_t, b_{t+1}, \ell_t, c_t^f, c_t^h} \sum_{t=0}^{\infty} \beta^t \left[ u(c_t^h, c_t^f) - v(\ell_t) \right],$$

$$\text{s.t.} \quad c_t^h + p c_t^f + \frac{b_{t+1}}{R^*} = b_t + \ell_t \left( 1 - \Upsilon \frac{\varphi}{2} \pi_t^2 \right)$$

$$\frac{1 - \omega}{\omega} \left( \frac{c_t^h}{c_t^f} \right)^{\frac{1}{\gamma}} = p (1 + \tau_t)$$

$$u_h(c_t^h, c_t^f) = \beta R^* u_h(c_{t+1}^h, c_{t+1}^f)$$

$$(1 + \pi_t) \pi_t = \frac{\varepsilon}{\varphi} \left[ \frac{v'(\ell_t)}{u_h(c_t^h, c_t^f)} - 1 \right] + \frac{\ell_{t+1}}{\ell_t} \frac{(1 + \pi_{t+1}) \pi_{t+1}}{R^*}$$

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s.t.  $c_t^h + p c_t^f + \frac{b_{t+1}}{R^*} = b_t + \ell_t$  ← Sticky prices induce costs only from output gap

$$\frac{1 - \omega}{\omega} \left( \frac{c_t^h}{c_t^f} \right)^{\frac{1}{\gamma}} = p (1 + \tau_t)$$

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Ramsey problem reduced to planner choosing  $\ell$  directly, while households choose  $c^h, c^f$

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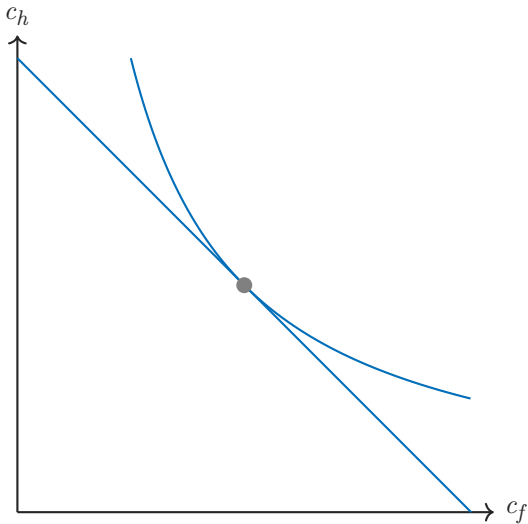
$$\frac{1 - \omega}{\omega} \left( \frac{c^h}{c^f} \right)^{\frac{1}{\gamma}} = p (1 + \tau)$$

Optimal policy:

$$\ell_t^{opt}(\tau) = \left( \frac{1+\tau}{1+\Theta_\tau^{-1}\tau} \right)^{\frac{\sigma}{1+\sigma\psi}} \left[ \frac{\Theta_\tau + \tau}{1+\tau} (\omega \Theta_\tau)^{\frac{\sigma-\gamma}{\gamma-1}} \right]^{\frac{1}{1+\sigma\psi}} > \ell_t^{look}(\tau).$$

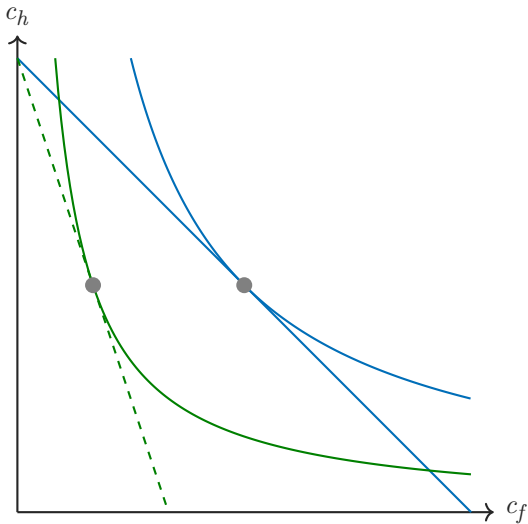
# Why is Stimulus Optimal?

$$\begin{aligned} \max_{c^h, c^f} \quad & u(c^h, c^f) - v(\ell) \\ \text{s.t.} \quad & c^h + p(1+\tau)c^f \leq w\ell + d + T \end{aligned}$$



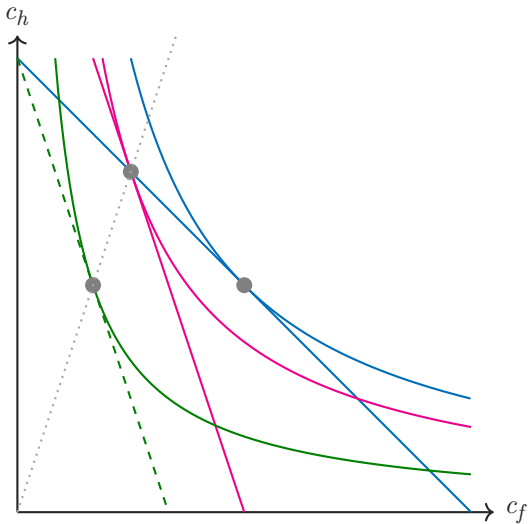
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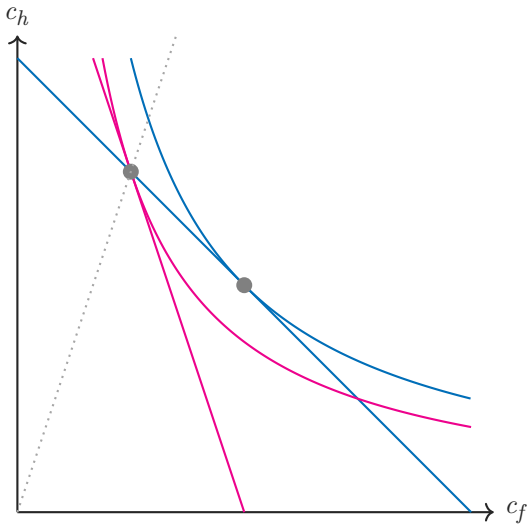
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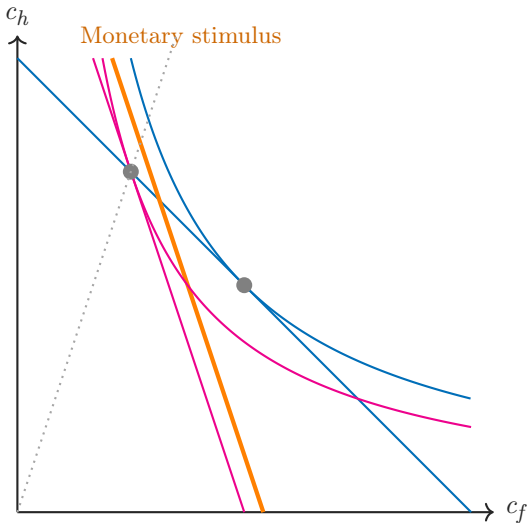
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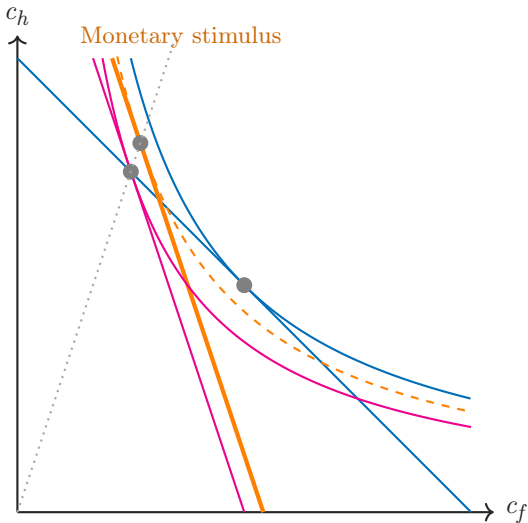
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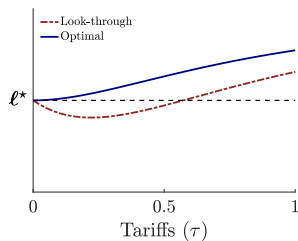
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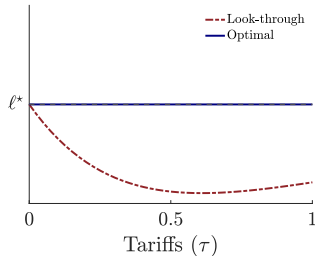


# Employment Response: Optimal Policy vs. Look-through

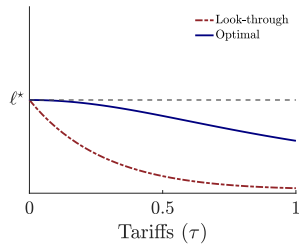
(a)  $\sigma = 0.5$



(b)  $\sigma = 1$

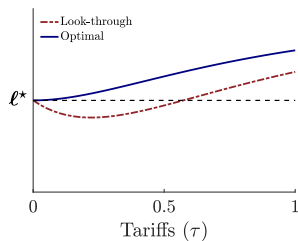


(c)  $\sigma = 2$

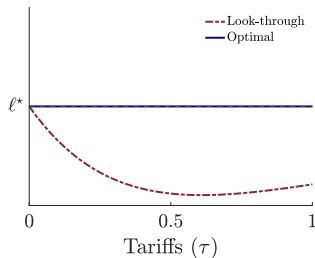


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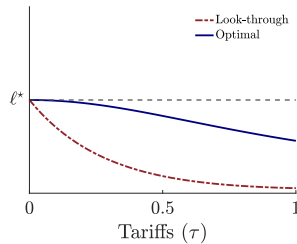
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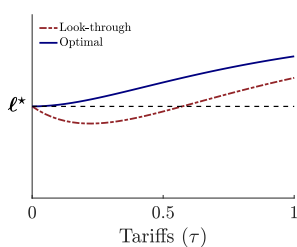
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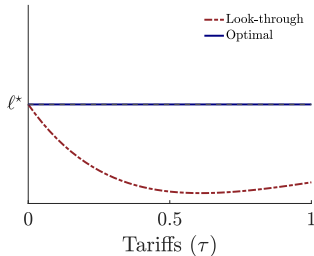
Under optimal policy, output is always above the natural level.

# Employment Response: Optimal Policy vs. Look-through

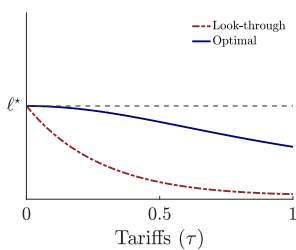
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(c)  $\sigma = 2$



Under optimal policy, output is always above the natural level.

- If  $\sigma < 1$ , output exceeds both natural and efficient level

↪ Unlike conventional cost-push shock

# Employment under Optimal Policy

Tariffs: Expansionary or Contractionary?

$$\frac{d \log \ell^{opt}}{d\tau} = \frac{(\Theta_\tau - 1)}{(1 + \sigma\psi)(1 + \tau)(\Theta_\tau + \tau)\Theta_\tau} (1 - \sigma)\gamma\tau$$

# Employment under Optimal Policy

Tariffs: Expansionary or Contractionary?

$$\frac{d \log \ell^{opt}}{d\tau} = \frac{(\Theta_\tau - 1)}{(1 + \sigma\psi)(1 + \tau)(\Theta_\tau + \tau)\Theta_\tau} (1 - \sigma)\gamma^\tau$$



No first-order effect on  $\ell$  at  $\tau = 0$

- At  $\tau = 0$ , planner purely rebalances  $c^h, c^f$

# Employment under Optimal Policy

Tariffs: Expansionary or Contractionary?

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- At  $\tau = 0$ , planner purely rebalances  $c^h, c^f$
- For  $\tau > 0$  the consumption distortion reduces the marginal return to labor leading to substitution and income effects
  - Employment response depends entirely on the IES
    - $\uparrow \ell \iff \sigma < 1$

# Quantitative Analysis

Standard NK assumption: price adjustment costs are not rebated,  $\Upsilon = 1$



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  - ▶ Starting from  $\pi = 0$ , costs of stimulating are second order, but first-order gains from mitigating fiscal externality
  - ▶ Stimulus only in the short-run  $\Leftarrow$  inflation in the long-run is too costly

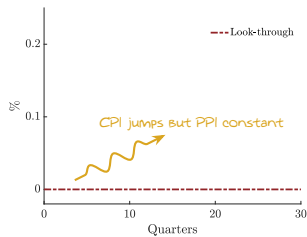
# Calibration

| Parameter     | Description                            | Value |
|---------------|----------------------------------------|-------|
| $\beta$       | Discount factor                        | 0.99  |
| $\gamma$      | Elasticity between $h$ and $f$         | 4     |
| $\sigma$      | IES                                    | 0.5   |
| $\psi$        | Inverse Frisch elasticity              | 1     |
| $\varepsilon$ | Elasticity of substitution (varieties) | 6     |

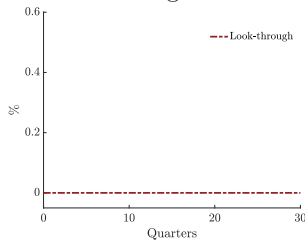
- Calibrate  $\varphi, \omega$  to match: (i) slope of Phillips Curve = 0.0055 (Hazell et al.); (ii) imports to tradable GDP
- Non-linear impulse response to permanent tariff  $\tau_t = 0.15$  (baseline)

# Permanent Tariff: Look-through

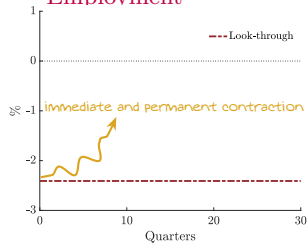
## Home-goods inflation



## Exchange rate



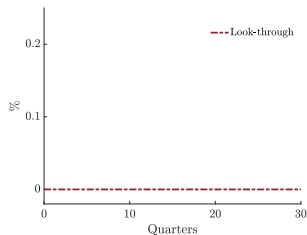
## Employment



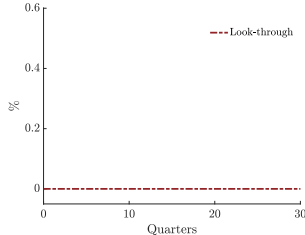
Inflation is annualized. Consumption, employment and the exchange rate are expressed in percentage deviation from the pre-tariff allocation. Trade balance are expressed as a fraction of GDP.

# Permanent Tariff: Look-through vs. Optimal Policy

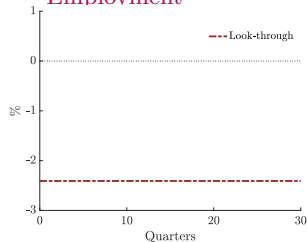
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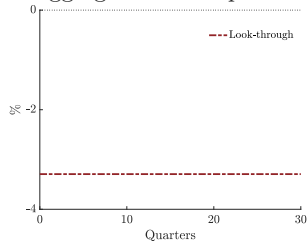
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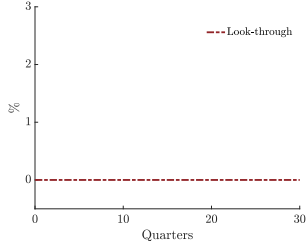
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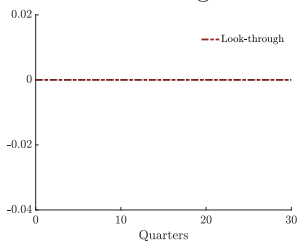
## Aggregate consumption



## Trade balance



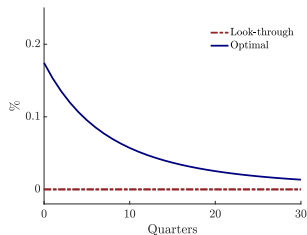
## Labor wedge



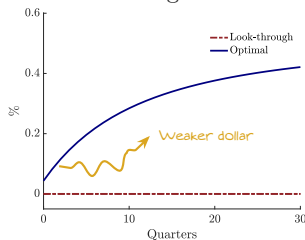
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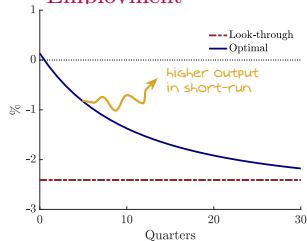
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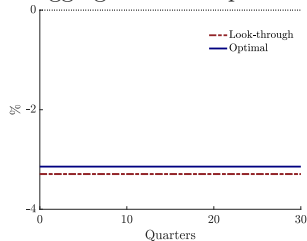
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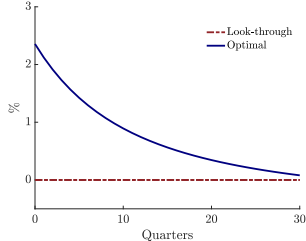
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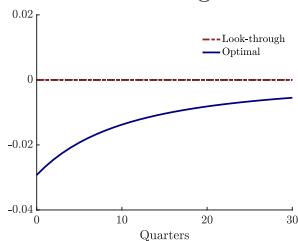
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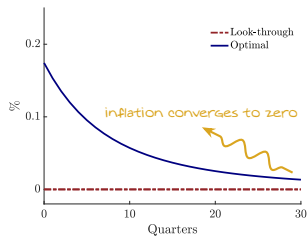


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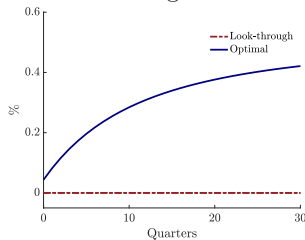


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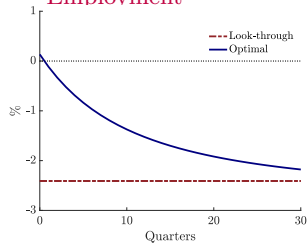
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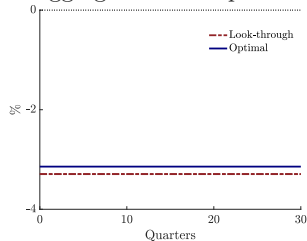
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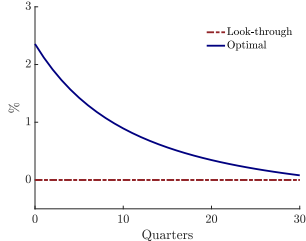
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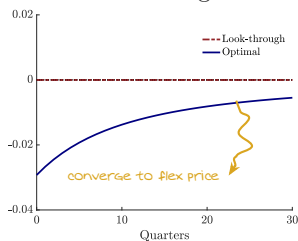
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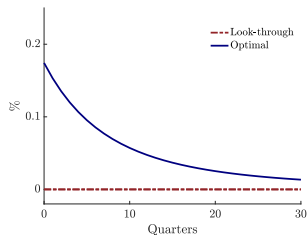
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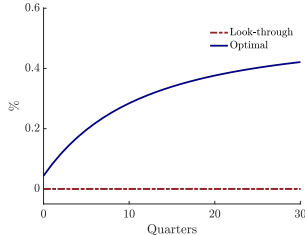
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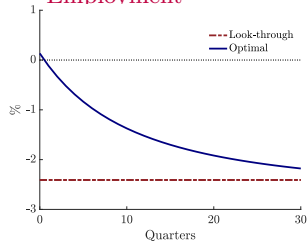
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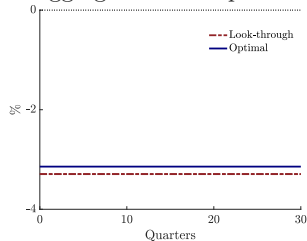
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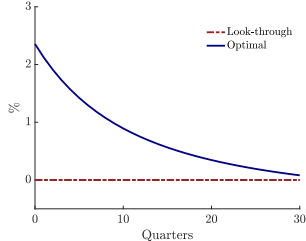
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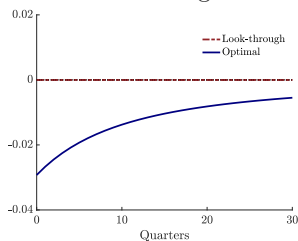
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# Taking Stock

- Optimal policy is to overheat the economy: inflation rises beyond *direct* effect of tariff
  - Increase in both imported and domestic produced goods
- Employment increases above both natural and efficient levels
- In the long-run, economy converges to flex-price allocation, with zero inflation, higher NFA position and lower employment

# Additional Results in the Paper

- Permanent shocks vs transitory   » Details
- Anticipated shocks:   » Details
  - ▶ Respond today, but less strongly
  - ▶ Larger trade deficit on impact
- PPI vs. CPI Targeting   » Details
  - ▶ Much larger losses under CPI targeting
- Main extensions
  - i) Endogenous terms-of-trade
  - ii) Imported intermediate inputs
  - iii) Distorted steady state
- Welfare

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## Endogenous TOT

- Continuum of SOE where  $c^f$  is a CES composite of goods produced abroad

$$c_{it} = \left[ \omega \left( c_{it}^h \right)^{1-\frac{1}{\gamma}} + (1-\omega) \left( c_{it}^f \right)^{1-\frac{1}{\gamma}} \right]^{\frac{\gamma}{\gamma-1}}, \quad c_{it}^f = \left( \int_0^1 \left( c_{it}^k \right)^{1-\frac{1}{\theta}} dk \right)^{\frac{\theta}{\theta-1}}$$

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- Export demand for home good

$$p_t = A(y_t - c_t^h)^{\frac{1}{\theta}} \quad \text{Baseline } \theta = \infty$$

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- Continuum of SOE where  $c^f$  is a CES composite of goods produced abroad

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► Quantitatively, modest attenuation of stimulus ► Results

# Tariffs on Imported Inputs

- Production of domestic varieties  $y_{jt} = \ell_{jt}^{1-\nu} x_{jt}^{\nu}$
- NK Phillips curve:

$$(1 + \pi_t)\pi_t = \frac{\varepsilon}{\varphi} [\textcolor{brown}{mc}_t - 1] + \beta \frac{u_h(c_{t+1}^h, c_{t+1}^f)}{u_h(c_t^h, c_t^f)} \frac{y_{t+1}}{y_t} (1 + \pi_{t+1})\pi_{t+1},$$

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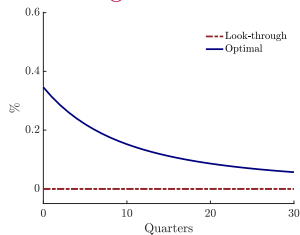
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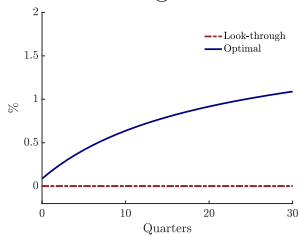
- Same as baseline: private cost of imported inputs exceeds social cost  
 $\Rightarrow$  Optimal policy is stimulative
- Quantitatively, larger increases in employment and welfare gains

# Tariff on Inputs Only

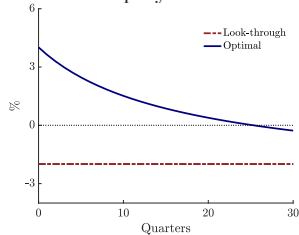
Home-goods inflation



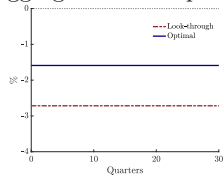
Exchange rate



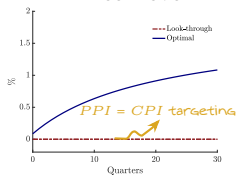
Employment



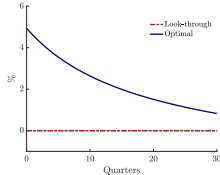
Aggregate consumption



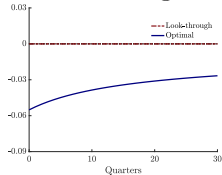
Price Level



Trade balance



Labor wedge



Calibrate  $\nu, \omega$  to match (i) share of intermediate inputs in total imports; (ii) imports-to-tradable GDP  
 » results w/ tariffs on inputs and consumption

# The case with distorted steady state

- Baseline model: labor subsidy is set to offset markup distortion

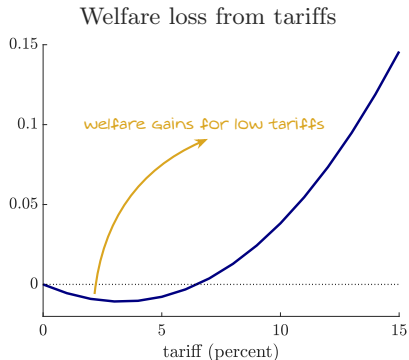
## The case with distorted steady state

- Start now from  $s = 0$  and use tariff revenue to subsidize labor  $P_t^f \tau_t c_t^f = s_t W_t \ell_t$ 
  - ▶ Unambiguous increase in employment
  - ▶ Inflation is mitigated despite larger increases in output



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# Welfare

|                                 | Gains Optimal Policy | Losses from Tariffs |              |
|---------------------------------|----------------------|---------------------|--------------|
|                                 |                      | Optimal Policy      | Look-through |
| <b>Baseline</b>                 | 0.009                | 0.99                | 1.00         |
| Anticipated tariffs             | 0.008                | 0.96                | 0.97         |
| Temporary tariffs               | 0.001                | 0.19                | 0.19         |
| Endogenous TOT                  | 0.007                | 0.68                | 0.69         |
| <b>Model w/ imported inputs</b> |                      |                     |              |
| Tariffs on $c$ and $x$          | 0.32                 | 1.61                | 1.91         |
| Tariffs on $c$                  | 0.01                 | 1.00                | 1.01         |
| Tariffs on $x$                  | 0.22                 | 0.59                | 0.80         |

*Note:* Welfare corresponds to permanent consumption equivalence (%).

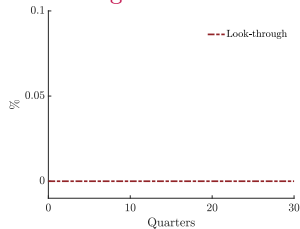
# Conclusions

- Optimal monetary policy response to tariffs is to overheat the economy
  - ▶ Monetary stimulus to offset fiscal externality
  - ▶ Let inflation rise above and beyond the direct effects from tariffs
- Reduction in trade deficit in response to permanent tariffs
- Dollar depreciation since April 2 is not puzzling

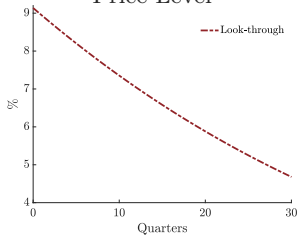
Extra Slides

# Temporary Tariff $\tau_t = 0.97 \cdot \tau_{t-1}$ ▶ back

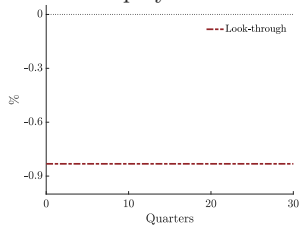
## Home-goods inflation



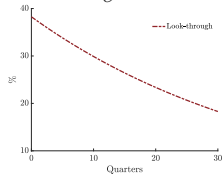
## Price Level



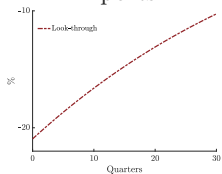
## Employment



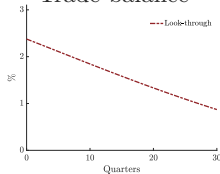
## $c^h$



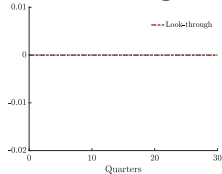
## Imports



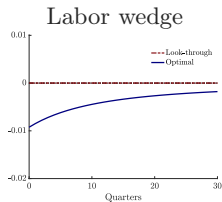
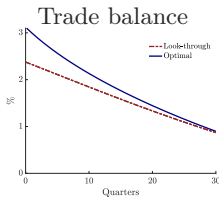
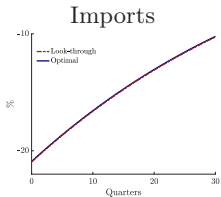
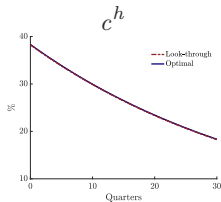
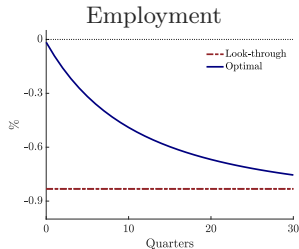
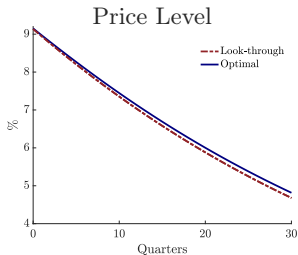
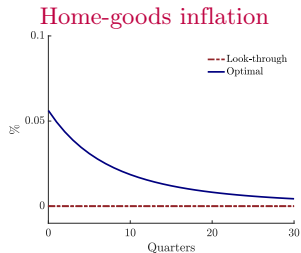
## Trade balance



## Labor wedge



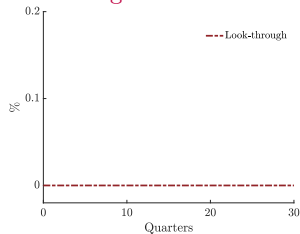
# Temporary Tariff $\tau_t = 0.97 \cdot \tau_{t-1}$ ▶ back



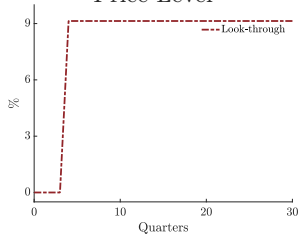
As in the case of a permanent tariff, optimal MP stimulates the economy

# Anticipation Effects [▸ back](#)

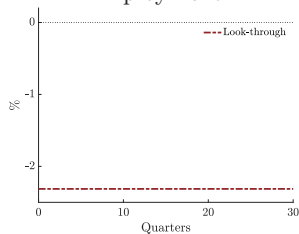
## Home-goods inflation



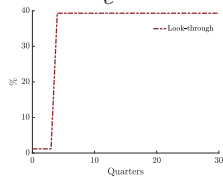
## Price Level



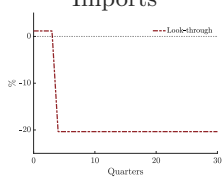
## Employment



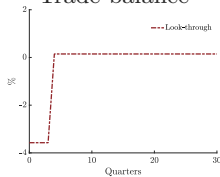
## $c^h$



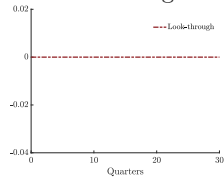
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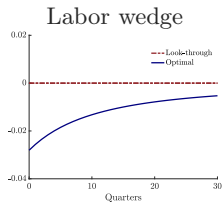
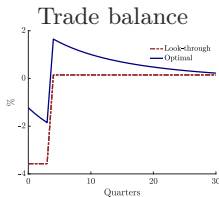
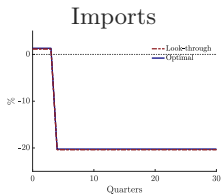
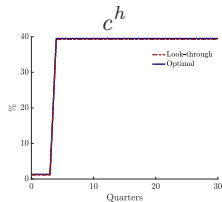
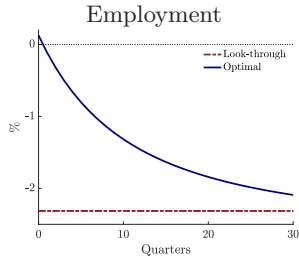
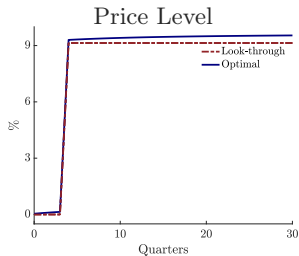
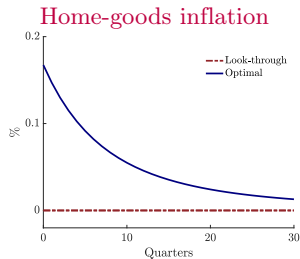
## Trade balance



## Labor wedge



# Anticipation Effects [▸ back](#)

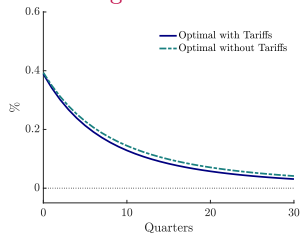


MP less expansionary: imports inefficiently high before tariff takes place

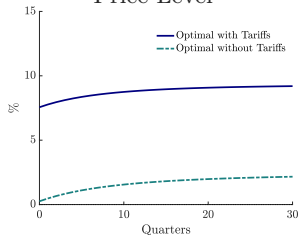


# The Case with Distorted Steady State [▸ back](#)

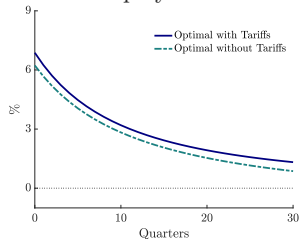
## Home-goods inflation



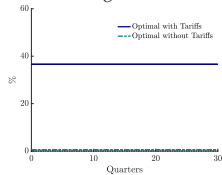
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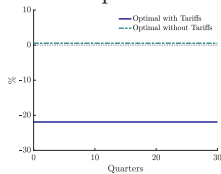
## Employment



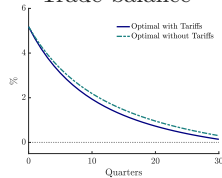
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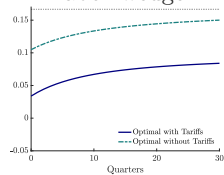
## Imports



## Trade balance



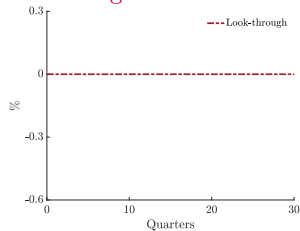
## Labor wedge



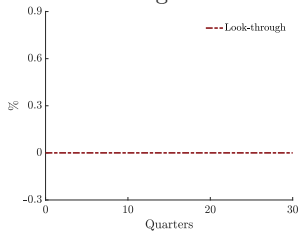
## CPI Targeting Rule

# Permanent Tariff » Back

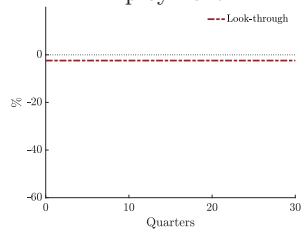
## Home-goods inflation



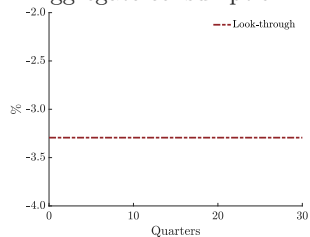
## Exchange rate



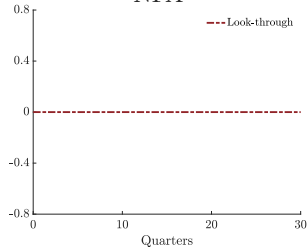
## Employment



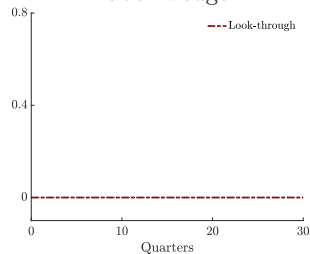
## Aggregate consumption



## NFA

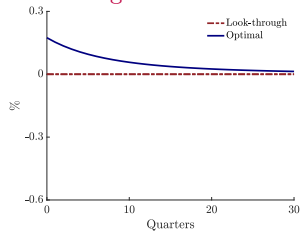


## Labor wedge

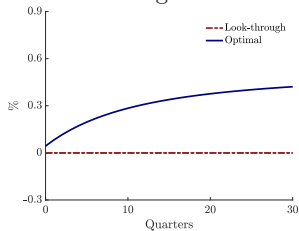


# Permanent Tariff » [Back](#)

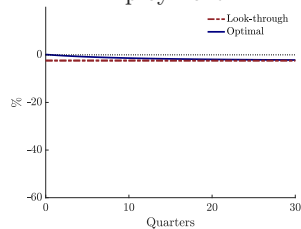
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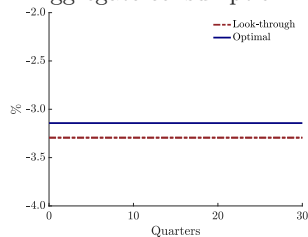
## Exchange rate



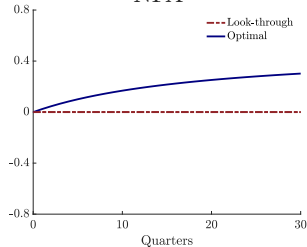
## Employment



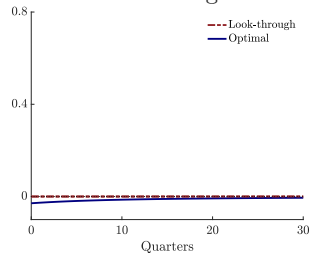
## Aggregate consumption



## NFA

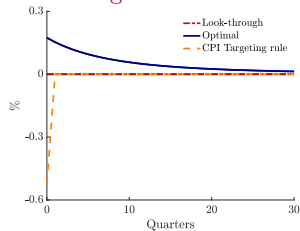


## Labor wedge

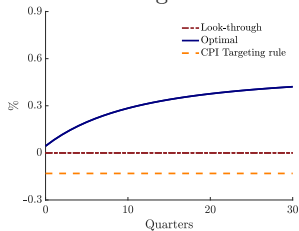


# Permanent Tariff [» Back](#)

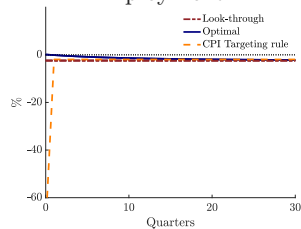
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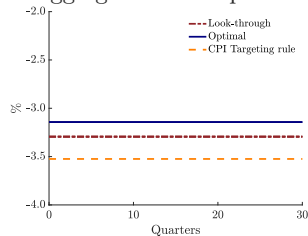
## Exchange rate



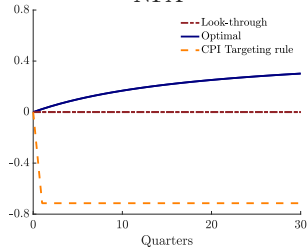
## Employment



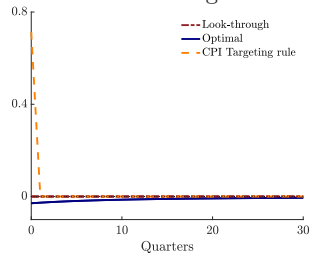
## Aggregate consumption



## NFA

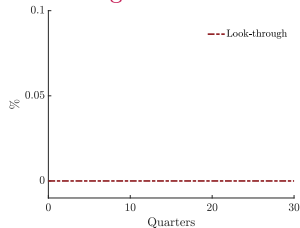


## Labor wedge

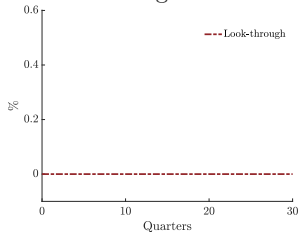


# Temporary Tariff $\tau_t = 0.97 \cdot \tau_{t-1}$ ▶ back

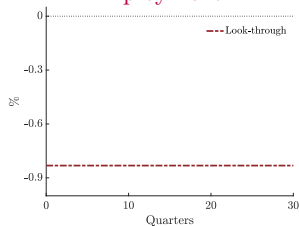
## Home-goods inflation



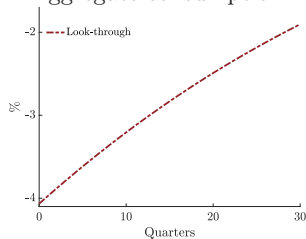
## Exchange rate



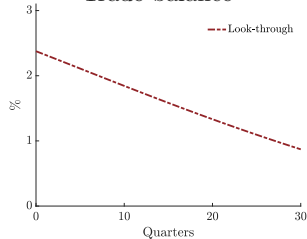
## Employment



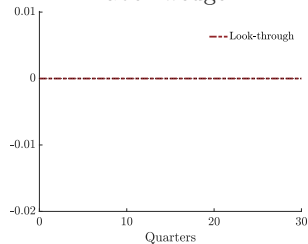
## Aggregate consumption



## Trade balance

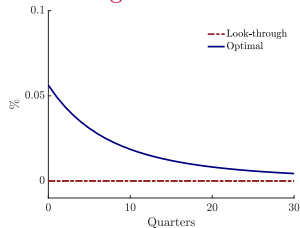


## Labor wedge

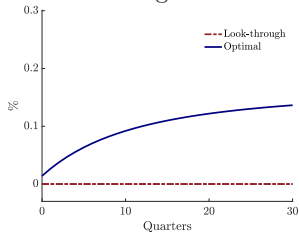


# Temporary Tariff $\tau_t = 0.97 \cdot \tau_{t-1}$ ▶ back

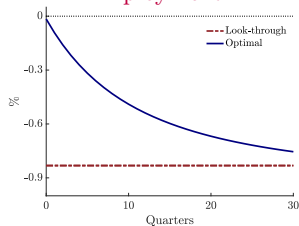
## Home-goods inflation



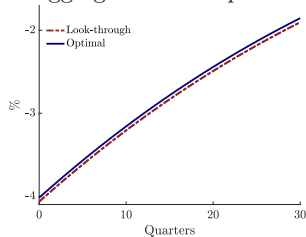
## Exchange rate



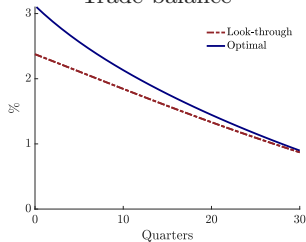
## Employment



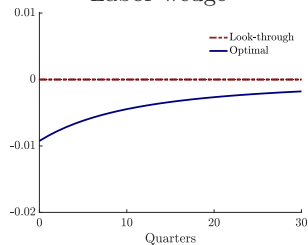
## Aggregate consumption



## Trade balance



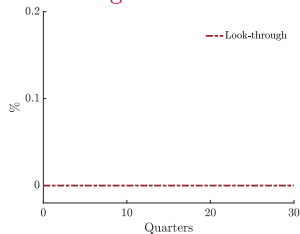
## Labor wedge



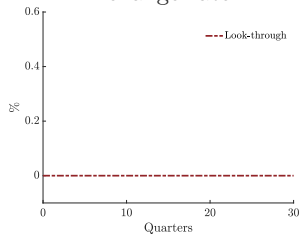
As in the case of a permanent tariff, optimal MP stimulates the economy

# Anticipation Effects [▶ back](#)

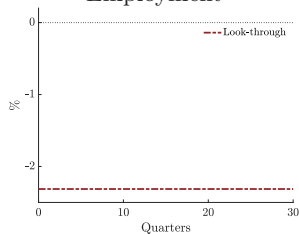
## Home-goods inflation



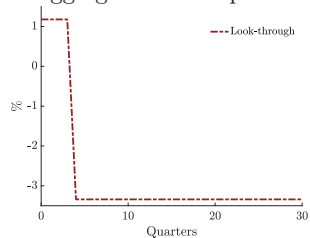
## Exchange rate



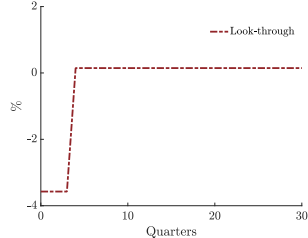
## Employment



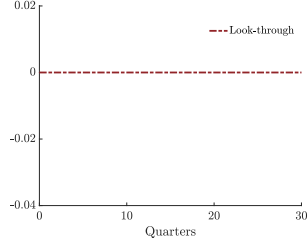
## Aggregate consumption



## Trade balance



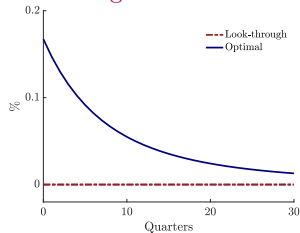
## Labor wedge



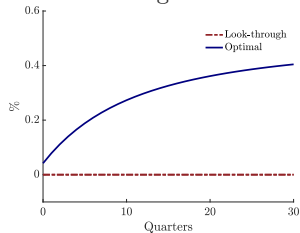


# Anticipation Effects [▶ back](#)

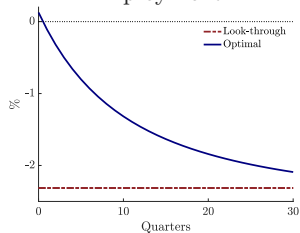
## Home-goods inflation



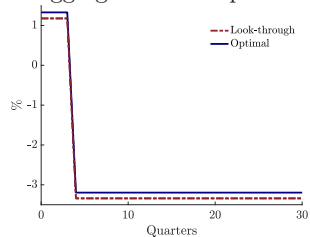
## Exchange rate



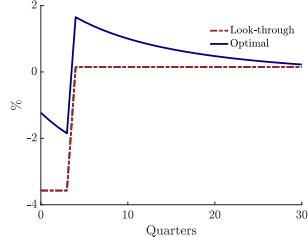
## Employment



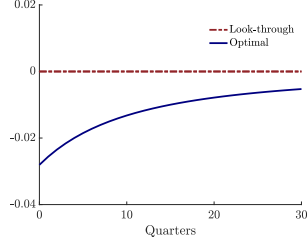
## Aggregate consumption



## Trade balance

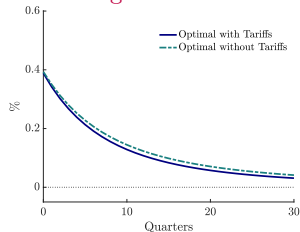


## Labor wedge

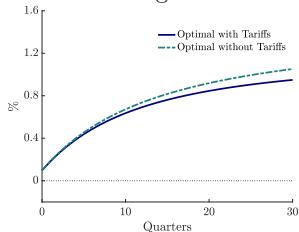


# The Case with Distorted Steady State [back](#)

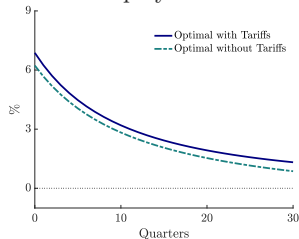
## Home-goods inflation



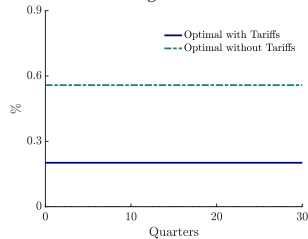
## Exchange rate



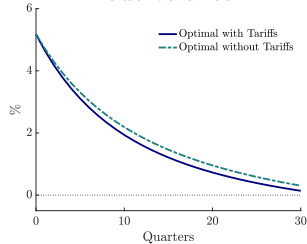
## Employment



## $c^h$



## Trade balance



## Labor wedge

